

Curriculum Vitae

Sumit Kumar Patra



Postdoctoral Researcher
Department of Chemistry,
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Toruń, Poland

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Career Objective

Motivated and passionate about working in the field of organic/Inorganic chemistry for research and development and taking on real-life challenges in the field of science. Having nine years of experience in organic and inorganic chemistry with an excellent track record of research success and multiple published international articles. Seeking an opportunity to be associated with a progressive organization or institution that gives me the scope to utilize and update my knowledge and research skills in the areas of organic/inorganic chemistry, chemical biology, and materials science in accordance with the latest trends.

Work Experience

➤ **Post-Doctoral Researcher: (April, 2025-Till Date)**

Department of Chemistry, Nicolaus Copernicus University, Toruń, Poland

Supervisor: Prof. Borys Osmialowski

Research Topic: The synthesis, spectral studies of new fluorescent dyes with variably oriented dipole moment

➤ **Post-Doctoral Researcher: (May, 2022-October, 2024)**

Chungnam National University (CNU), Daejeon, South Korea

Supervisor: Prof. Young A. Son

Research Topic:

- Fluorescence probes for bio sensing and environmental pollutant
- Development and Photophysical study several dual state emissive materials.
- Modification of Lignin polymer and increase the hydrophobic nature.

➤ **Scientific Officer: (January 2022- April 2022)**

North Eastern Hill University, India

High-resolution mass spectrometry (HRMS) (Waters, Model: XeVo G2-XS QTof),

Academic Qualification

- **PhD in Chemistry: (2014-2021)**

Department of Chemistry

North Eastern Hill University, Shillong, India,

Thesis title: Development of Photoluminescent Organic Probe and Ruthenium(II) Complexes for Multianalyte Sensing and Bioimaging.

Supervisor: Dr. Snehadrinarayan Khatua

Description: This thesis explores on design and synthesis of new organic and Ruthenium compound for detection and quantification of environmentally hazardous and biologically important ion or molecule, analytical analysis and computational study.

- **Master of Science in Chemistry: (2012-2014)**

Vidyasagar University, West Bengal, India

82.33%, First Class.

- **Bachelor of Science in Chemistry honours: (2009-2012)**

Vidyasagar University, West Bengal, India

69.75%, First Class.

- **10+2 in Science, 2009**

West Bengal Council of Higher Secondary Education, India,

81.40%, First Class.

- **10th standard, 2007,**

West Bengal Board of Secondary Education, India,

86.00%, First Class.

Research Publication

22. Multi-state efficient luminogens with solvatochromic features based on a conjugation induced twisted framework for highly selective dual channel detection of mercury ion.

Sumit Kumar Patra, Ramalingam Manivannan, Young-A Son

[*Dyes and Pigments*, **2025**, 233, 112534](#)

21. An organic emitter with enhanced fluorescence modulation, high emission efficiency, multicolor tunability in solution and solid states, and dual-channel sensing for selective cyanide detection.

Ramalingam Manivannan, **Sumit Kumar Patra**, Palanisamy Jayasudha, Young-A Son.
[*Spectrochim Acta A Mol Biomol Spectrosc.*, **2025**, 335, 125922.](#)

20. A triple-chromophore based NIR fluorescent chemodosimeter with configurable opto-chemical logic gate functions for multi-analyte recognition of N₂H₄, CN[−], and ClO[−] in real samples.

Gaurav Bartwal, Ramalingam Manivannan, **Sumit Kumar Patra**, Yoonwoo Choi, Young-A. Son.

[*Spectrochim Acta A Mol Biomol Spectrosc.*, **2025**, 339, 126250](#)

19. A Dual-State Emissive TICT-AIEE-active Near-Infrared Luminogen with Multi-Stimuli-Responsive Chromism for Information display and Data Encryption.

Gaurav Bartwal, Ramalingam Manivannan, **Sumit Kumar Patra**, Young-A. Son.

[*Dyes and Pigments*, **2025**, 112870, DOI: 10.1016/j.dyepig.2025.112870](#)

18. Stability and photophysical study of bright day light fluorescent material and its fabrication in laser patterning and LED application.

Ramalingam Manivannan, Kaliyan Prabakaran, **Sumit Kumar Patra**, Hyunseo Kim, Sang Kyoo Lim, Young-A Son

[*Dyes and Pigments*, **2025**, 232, 112463](#)

17. A dual state emission luminogen based on 1,3,3-trimethylindoline and chroman-2,4-dione conjugate for highly selective dual channel detection of cyanide ion.

Snehadrinarayan Khatua*, **Sumit Kumar Patra**, Monosh Rabha, Deikrisha Lyngdoh Lyngkhoi, Jogat Gogoi, Bhaskar Sen.

[*Sens. Diagn.*, **2024**, 3, 1689-1698](#)

16. Cyclometalated Iridium(III) Polypyridine Complexes for Luminescent Detection of Chlorinated Solvents.

Nimisha Kashyap, Monosh Rabha, **Sumit Kumar Patra**, Bhaskar Sen, Sanjoy Kumar Sheet, Khanindram Baruah, and Snehadrinarayan Khatua.

[*Cryst. Growth Des.* **2024**, *24*, 3615–3631](#)

15. Tailored fluorophore design: Enhancing selectivity for cyanide ion sensing in water and food samples, and innovative device development.

Palanisamy Jayasudha, Ramalingam Manivannan, Wonbin Kim, Hyeongcheon Lim, **Sumit Kumar Patra**, Young-A. Son.

[*Spectrochim Acta A Mol Biomol Spectrosc.* **2024**, *314*, 124196](#)

14. AIE-active cyclometalated iridium(III) complexes for the highly efficient picric acid detection in aqueous media.

Monosh Rabha, Deikrisha Lyngdoh Lyngkhai, **Sumit Kumar Patra**, Snehadrinarayan Khatua.

[*Inorg. Chem. Front.*, **2024**, *11*, 1434-1449](#)

13. Effect of thermal, mechanical and photophysical properties of high emissive photoluminescence organic pigments.

Ramalingam Manivannan, Minwoo Han, **Sumit Kumar Patra**, Kaliyan Prabakaran, Wonbin Kim, Sang Kyoo Lim, Jiyeon Oh, Young-A Son

[*Dyes and Pigments*, **2024**, *221*, 111785](#)

12. Multicolor emissive organic material to display aggregation caused red shift with dual state emission, and application towards rewritable data storage.

Sumit Kumar Patra, Ramalingam Manivannan, Young-A Son

[*J. Photochemistry. Photobiology. A*, **2023**, *444*, 114945](#)

11. Aggregation induced emission active bis-heteroleptic ruthenium(II) complex for luminescent light-up detection of pyrophosphate ion.

Sumit Kumar Patra, Bhaskar Sen, Monosh Rabha, Kripamoy Aguan and Snehadrinarayan Khatua

[*Dalton Trans.*, **2023**, *2023*, 2592-2602](#)

10. Aggregation-induced emission-active bis-heteroleptic ruthenium(II) complex of thiophenyl substituted phenanthroline for selective “turn-off” detection of picric acid.

Sumit Kumar Patra, Bhaskar Sen, Monosh Rabha and Snehadrinarayan Khatua

[*New J. Chem.*, **2022**, 46, 169-177.](#)

9. Ruthenium(II) Complex-Based AIEgen for Rapid and Selective Luminescent Detection of Phosgene in Solution and Gas Phase.

Bhaskar Sen, **Sumit Kumar Patra**, and Snehadrinarayan Khatua

[*Inorg. Chem.* **2021**, 60, 19175–19188.](#)

8. Visible-light-mediated synthesis of 3,4,5-trisubstituted furan-2-one derivative via bifunctional based organo photocatalyst.

Arup Dutta, **Sumit Kumar Patra**, Snehadrinarayan Khatua and Rishanlang Nongkhaw

[*New J. Chem.*, **2021**, 45, 19329-19337.](#)

7. Luminescent Detection of Ag⁺ and Phosphate Ions by a Ruthenium(II) Complex-based Multianalyte Probe: A Combined Spectroscopic, Crystallographic and Theoretical Approach.

Bhaskar Sen, **Sumit Kumar Patra**, Monosh Rabha, Sanjoy K. Sheet, Kripamoy Aguan, Debabrata Samanta, Snehadrinarayan Khatua

[*Eur. J. Inorg. Chem.*, **2021**, 3549–3560.](#)

6. Ruthenium(II) Complex-based G-quadruplex DNA Selective Luminescent ‘Light-up’ Probe for RNase H Activity Detection.

Sanjoy Kumar Sheet, Monosh Rabha, Bhaskar Sen, **Sumit Kumar Patra**, Kripamoy Aguan, Snehadrinarayan Khatua.

[*ChemBioChem*, **2021**, 22, 2880-2887.](#)

5. Highly Selective Detection of Hypochlorous Acid by a Bisheteroleptic Ru(II) Complex of Pyridyl-1,2,3-triazole Ligand via C(sp²)-H Hydroxylation.

Bhaskar Sen, Sanjoy Kumar Sheet, **Sumit Kumar Patra**, Debaprasad Koner Nirmalendu Saha, Snehadrinarayan Khatua.

[*Inorg. Chem.* **2019**, 58, 9982–9991.](#)

4. Single Molecular Dual Analyte Thio-Urea based Probes for Colorimetric Hg²⁺ and Fluorometric AcO[−] Detection and Its’ Application in Bioimaging.

Sumit Kumar Patra, Bhaskar Sen, Sanjoy Kumar Sheet, Bodhisattwa Banerjee, Nirmalendu Saha, Snehadrinarayan Khatua.

[*Inorganica Chimica Acta*, **2019**, 492, 119-130.](#)

3. Aggregation-Induced Emission-Active Ruthenium(II) Complex of 4,7-Dichloro Phenanthroline for Selective Luminescent Detection and Ribosomal RNA Imaging.

Sanjoy Kumar Sheet, Bhaskar Sen, **Sumit Kumar Patra**, Monosh Rabha, Kripamoy Aguan, Snehadrinarayan Khatua.

[*ACS Appl. Mater. Interfaces*, **2018**, *10*, 14356–14366.](#)

2. Highly Sensitive Bifunctional Probe for Colorimetric Cyanide and Fluorometric H₂S Detection and Bioimaging: Spontaneous Resolution, Aggregation, and Multicolor Fluorescence of Bisulfide Adduct.

Sumit Kumar Patra, Sanjoy Kumar Sheet, Bhaskar Sen, Kripamoy Aguan, Debesh Ranjan Roy, Snehadrinarayan Khatua.

[*J. Org. Chem.*, **2017**, *82*, 10234–10246.](#) (Top most downloaded article during Sept.-Oct., 2017)

1. Study of Antioxidant Property of the Pseudo Bulb Extract of *Crepidium Acuminatum* (Jeevak) and its Use in the Synthesis of Gold Nanoparticles.

Braja Gopal Bag*, Shib Shankar Dash, **Sumit Kumar Patra**.

[*Int. J. Res. Chem. Environ*, **2014**, *4*, 133-138](#)

Research Experience

- Specialized hands-on expertise in design and multistep synthesis of fluorescent material and Ru(II) complex.
- Familiar in organic/ inorganic photochemistry.
- Broad experience in development of sensors/probes for cation, anion, toxic analyte, chemical warfare agent and biologically important species
- Experience in dual state emission, aggregation induced emission and self-assembly.
- Extensive experience in design and developing of highly emissive luminescence organic pigments and xanthene based color filter materials.
- Experience in hydrophobic modification of lignin polymer and details mechanical study.

Technical Experience

1. Hand on experience in handling **HRMS** (Waters, Xevo G2-XS QToF) as an operator at Dept. of Chemistry, NEHU, Shillong, India.

2. Extensive experience in handling **NMR** (Bruker Advance, 400 MHz) Spectrometer as an operator at SAIF, NEHU, Shillong, India.
3. Hands-on experience on **FT-IR** (Perkin-Elmer) instrument.
4. Experience in handling **UV-Visible spectrophotometer** (Perkin-Elmer Lambda 25 UV-vis scanning spectrophotometer; Carry 60 UV-vis spectrophotometer, Agilent Technologies)
5. Experience in handling **Fluorescence** (Hitachi F-4500 and Perkin-Elmer LS55, Varian carry Eclipse) Spectro fluorophotometer.
6. Experience in handling **Cyclic Voltammetric instrument** (CH Instruments and Metrohm instrument, Model No. Metrohm Autolab M204).
7. Good experience on *in vitro* and *in vivo* cell imaging study.
8. Knowledge on Single Crystal X-Ray data collection and structure solution.
9. Good experience on singlet and triplet state DFT and TD-DFT calculation using density functional theory (DFT) on Gaussian 09 and Gaussian 16 program.

Use of Software

1. Experience in using **MestRec** and **MestReNova** software, for 1D and 2D NMR spectral data (^1H , ^{13}C , ^1H - ^1H COSY, ^1H - ^{13}C HSQC, ^1H - ^{13}C HMBC etc) processing and visualization.
2. Experience in using **MestReNova** software for interpretation of Mass data.
3. Experience in using **Origin** software for graph plot and spectral data fitting.
4. Experience in using **Isopro** for simulation of mass spectra.
5. Experience in using **HyperChem** and **gaussian** software for theoretical calculation.
6. Experience in using **Chem. Draw** for structure drawing.
7. Experience in using photographic software like **Adobe Photoshop CS3, GIMP, and paint.**

Academic Achievement

- ❖ Graduate Aptitude Test in Engineering (GATE) 2014, 2015 conducted jointly by IISc and IITs on behalf of MHRD, Government of India
 - GATE Rank- **631**, with a GATE score 433, 2014
 - GATE Rank- **1212**, with a GATE score 417, 2015

- ❖ **CSIR-SRF (Direct)** for Research Fellowship under CSIR Fellowship Scheme] in the year 2018.

Conference Attended

1. **Sumit Kr. Patra** and Snehadrinarayan Khatua, **Poster presentation**. National Seminar on Contemporary Challenges in Chemistry (**NSC3-2018**), NEHU, Shillong, 20th-21st November, 2018.
2. **Sumit Kr. Patra** and Snehadrinarayan Khatua, **Poster presentation**. International Conference on Chemistry for Human Development (**ICCHD-2018**), Heritage Institute of Technology, Kolkata, 8th to 10th January, 2018.
3. **Sumit Kr. Patra** and Snehadrinarayan Khatua, **Poster presentation**. Recent Trends in Chemical Science (**RTCS-2017**), National Institute of Technology Meghalaya, Shillong, 12th -13th October, 2017.

Additional Experience during PhD and Postdoctoral Program

- Mentoring postgraduate students in their project in Dr. Khatua's Research Group, North eastern Hill University, INDIA.
- Advising graduate students in the Prof. Son's Research Group, Chungnam National University (CNU), South Korea.

Personal Details

Name:	Sumit Kumar Patra
Date of Birth:	9 th December, 1990
Nationality:	Indian
Gender:	Male
Marital Status:	Married
Languages:	English, Bengali (mother tongue), Hindi
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References

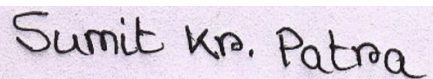
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Declaration:

I, hereby declare that the above-furnished particulars are true to the best of my knowledge and belief. If given a chance, I will prove my efficiency, my loyalty and willingness to work.



Sumit Kr. Patra